Akoizo Original Practice - 2027 Rotation Form 1

Division C | Anatomy And Physiology | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Aspirated peanut most likely lodges in which bronchus and why?

Unit: Airways: From Nose to Bronchioles.
Worked-case data: Tube A has the baseline radius. Tube B has 80% of that radius. Both have the same length, viscosity, and driving pressure.

A. Right, wider and more vertical

B. Left, more horizontal

C. Trachea only

D. Left, narrower

Answer: _____

2. (2 points) Vital capacity equals:

Unit: Ventilation, Volumes, and Spirometry.
Worked-case data: Person A breathes 500 mL twelve times each minute. Person B breathes 250 mL twenty-four times. Both have 150 mL anatomical dead space per breath.

A. ERV + RV

B. TV + RV

C. TV + IRV

D. IRV + TV + ERV

Answer: _____

3. (2 points) Reversible obstruction with eosinophils best fits:

Unit: Obstructive Disease: Asthma, Bronchitis, Emphysema, COPD.
Worked-case data: Dataset A has variable wheeze, reduced expiratory ratio, and largely preserved diffusion capacity. Dataset B has persistent obstruction, increased residual volume, and reduced diffusion capacity.

A. Pneumothorax

B. Emphysema

C. Asthma

D. Fibrosis

Answer: _____

4. (2 points) Pepsinogen is converted to pepsin by:

Unit: Alimentary Canal: Mouth to Stomach.
Worked-case data: An educational case describes loss of parietal-cell function with adequate dietary B12 but poor B12 absorption.
Separate investigation observations:
Diet record: The case provides adequate vitamin B12 intake.
Gastric finding: Parietal-cell function is greatly reduced.
Distal finding: The terminal ileum is anatomically present, but normal B12 uptake is impaired.

A. Bile

B. HCl/acid pH

C. Bicarbonate

D. Insulin

Answer: _____

5. (2 points) Duodenal ulcer pain classically:

Unit: Ulcers, GI Bleeding, and GI Cancers.
Worked-case data: Case A supplies impaired mucus protection and a focal mucosal defect. Case B supplies an invasive colonic lesion that narrows the lumen.
Separate investigation observations:
Mucosal image description: A focal defect extends through the mucosa.
Protective function: Mucus-bicarbonate protection is reduced.
Lumen assessment: No obstructing tissue mass is described.

A. Better with food/antacid

B. No relation

C. Only nocturnal

D. Worse with food

Answer: _____

6. (2 points) B cells mature in the ___; T cells mature in the ___.

Unit: Central Immune Organs and Lymph Flow.
Worked-case data: A small skin abrasion introduces material into tissue fluid of the left foot. Trace a plausible route toward immune surveillance and return to blood.
Separate investigation observations:
Origin: Antigen enters interstitial fluid around a skin abrasion.
Transport: Afferent lymph carries antigen-bearing cells from the tissue.
Cell encounter: Mature naive lymphocytes meet presented antigen in organized lymphoid tissue.

A. Spleen; node

B. Node; spleen

C. Thymus; marrow

D. Marrow; thymus

Answer: _____

7. (2 points) MAC is formed by:

Unit: Innate Immunity and Complement.
Worked-case data: A hypothetical dataset shows poor complement-mediated coating of microbes, while antibody production is present. A second dataset shows coating but failure of terminal membrane attack.
Separate investigation observations:
Recognition assay: Microbial targets are recognized.
Coating assay: C3b deposition on targets is markedly reduced.
Phagocyte assay: Phagocytes work better when an external opsonin is supplied.

A. Histamine

B. C3b only

C. C5b-9

D. IgE alone

Answer: _____

8. (2 points) AIDS is defined by CD4 under ___ or opportunistic illness.

Unit: Immunodeficiency: HIV, SCID, and CVID.
Worked-case data: A hypothetical case has circulating B cells but repeatedly poor antibody responses to supplied antigen challenges. T-cell measurements are also provided and appear preserved.
Separate investigation observations:
Cell inventory: B cells are present in the sample.
Challenge response: Antibody output after repeated specified challenges is poor.
Comparison: The supplied T-cell functional comparison is preserved.

A. 500

B. 350

C. 200

D. 50

Answer: _____

9. (4 points) Which cells and structures drive mucociliary clearance?

Unit: Airways: From Nose to Bronchioles.
Worked-case data: Tube A has the baseline radius. Tube B has 80% of that radius. Both have the same length, viscosity, and driving pressure.

Answer: _____
_____
_____

10. (2 points) FEV1/FVC 0.55 with reduced FEV1 suggests:

Unit: Ventilation, Volumes, and Spirometry.
Worked-case data: Person A breathes 500 mL twelve times each minute. Person B breathes 250 mL twenty-four times. Both have 150 mL anatomical dead space per breath.

A. Normal

B. Mixed only

C. Restriction

D. Obstruction

Answer: _____

11. (2 points) A teaching spirometry case gives tidal volume 520 mL, dead space 150 mL and respiratory rate 12/min. Find alveolar ventilation.

Answer: _____
_____
_____

12. (2 points) Use the same case to calculate minute ventilation: tidal volume 520 mL and rate 12/min.

Answer: _____
_____
_____

13. (2 points) A model narrows an airway while pressure difference and other conditions are fixed. Explain the likely airflow consequence.

Answer: _____
_____
_____

14. (2 points) Describe how pleural fluid assists ordinary breathing.

Answer: _____
_____
_____

15. (2 points) Why does the small intestine combine villi, microvilli and a thin absorptive epithelium?

Answer: _____
_____
_____

16. (2 points) Distinguish the digestive roles of bile and pancreatic lipase.

Answer: _____
_____
_____

17. (2 points) Explain why a pancreatic bicarbonate secretion can assist digestion of acidic stomach contents.

Answer: _____
_____
_____

18. (2 points) Compare innate and adaptive immune responses without calling either one universally faster or stronger in every situation.

Answer: _____
_____
_____

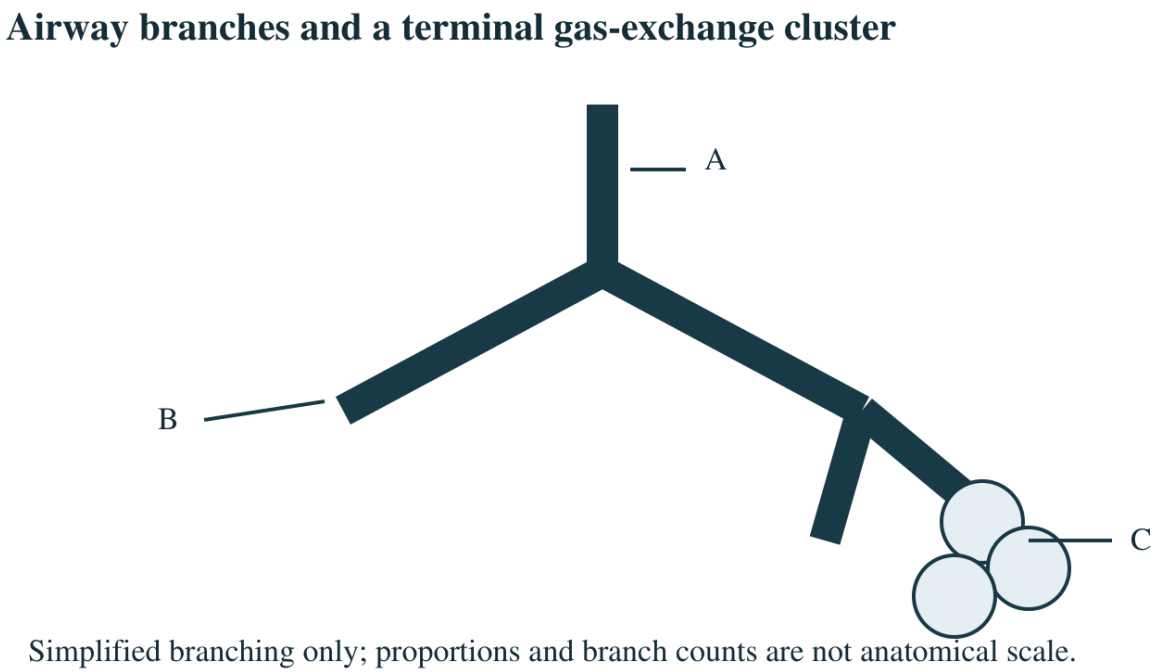
19. (2 points) Explain why a secondary antibody response can differ from a primary response.

Answer: _____
_____
_____

20. (2 points) In a teaching case, helper T-cell activation is impaired. Give one potential consequence for a coordinated adaptive response.

Answer: _____
_____
_____

21. (2 points) Use Figure F1. Name the structure labeled A.



F1. Respiratory branching schematic (original synthetic teaching figure)

Answer: _____
_____
_____

22. (2 points) Use Figure F1. Which labeled region most directly provides the thin gas-exchange interface?

Refer to F1. Respiratory branching schematic (original synthetic teaching figure) shown with the first question in this figure group.

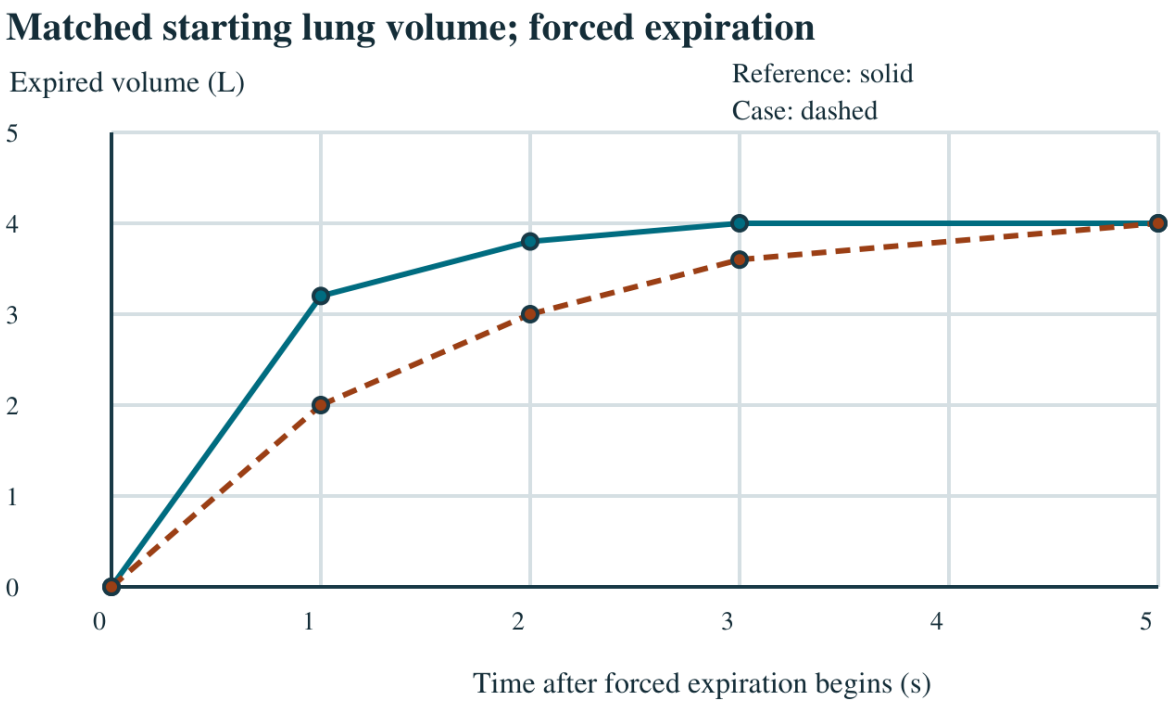
Answer: _____
_____
_____

23. (2 points) Use Figure F1. Trace inspired air from A toward C using the branching arrangement.

Refer to F1. Respiratory branching schematic (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

24. (2 points) Use Figure F2. Calculate case FEV1/FVC as a percentage.



F2. Forced expiratory volume curves (original synthetic teaching figure)

Answer: _____
_____
_____

25. (2 points) Use Figure F2. Which curve shows slower early emptying, and what pattern does that support studying?

Refer to F2. Forced expiratory volume curves (original synthetic teaching figure) shown with the first question in this figure group.

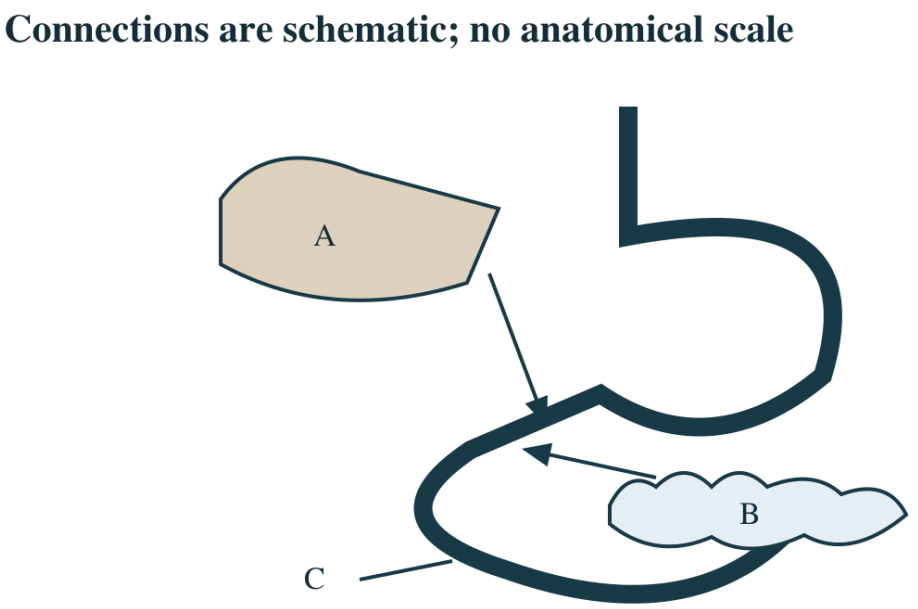
Answer: _____
_____
_____

26. (2 points) Use Figure F2. Find reference-minus-case expired volume at1 s.

Refer to F2. Forced expiratory volume curves (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

27. (2 points) Use Figure F3. Which labeled organ produces bile? Give its name.



A large accessory organ; B lobulated gland; C coiled intestinal region

F3. Digestive organ relationships (original synthetic teaching figure)

Answer: _____
_____
_____

28. (2 points) Use Figure F3. Name B and explain one digestive secretion it delivers toward the shown intestinal connection.

Refer to F3. Digestive organ relationships (original synthetic teaching figure) shown with the first question in this figure group.

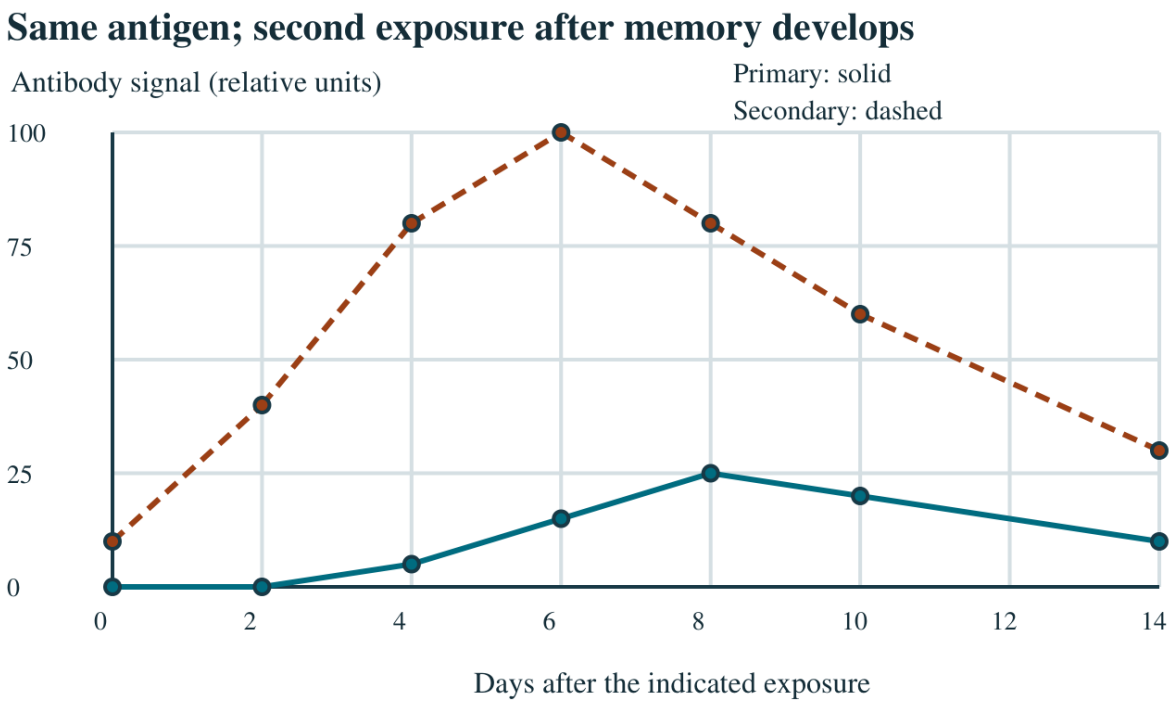
Answer: _____
_____
_____

29. (2 points) Use Figure F3. Which labeled region is most directly associated with most nutrient absorption?

Refer to F3. Digestive organ relationships (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

30. (2 points) Use Figure F4. Read the ratio secondary-peak / primary-peak signal.



F4. Primary and secondary antibody responses (original synthetic teaching figure)

Answer: _____
_____
_____

31. (2 points) Use Figure F4. At day4, calculate secondary-minus-primary signal.

Refer to F4. Primary and secondary antibody responses (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

32. (2 points) Use Figure F4. Explain how the curve comparison is consistent with adaptive immune memory.

Refer to F4. Primary and secondary antibody responses (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____